

4. [Maximum mark: 8]

For $x \in \mathbb{R}$, the functions f and g are defined by

$$f(x) = -x^2 + 4x + p \text{ and}$$

$$g(x) = x^2 + qx - 1,$$

where p and q are non-zero constants.

The graph of $y = f(x)$ has a vertex at A and the graph of $y = g(x)$ has a vertex at B.

Points A and B both lie on the line $y = 2x - 1$.

(a) Show that $p = -1$. [3]

(b) Find the value of q . [5]

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